Answer Key — Keeping Time: Day, Month, Year and Calendars

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) rotation on its axis** — Earth's rotation gives the day (~24 hours).

Q2 **(b) Moon's cycle of phases** — the ~29.5-day phase cycle of the Moon gives the month.

Q3 **(c) 365¼ days** — Earth's revolution takes about 365¼ days.

Q4 **(b) at midday (Sun highest)** — the shadow is shortest when the Sun is highest, at midday.

Q5 **(a) Both true, R explains A** — the extra quarter-days make a leap day every 4 years — as R explains.

Q6 **lunar** — phase-based calendars are lunar calendars.

Q7 **sundial** — a sundial uses the Sun's shadow.

Short answer

Q8 Day: Earth's rotation on its axis; Month: Moon's cycle of phases; Year: Earth's revolution around the Sun (cycle of seasons).

Q9 Mark the shadow tip through the day; the shortest shadow occurs when the Sun is highest, which is local noon (midday).

Q10 A lunar calendar is based on the Moon's phase cycle (~12 lunar months per year); a solar calendar is based on Earth's revolution around the Sun (the year).

Long / application

Q11 Earth's rotation on its axis gives the day (~24 hours); the Moon's cycle of phases as it revolves around Earth gives the month (~29.5 days); Earth's revolution around the Sun, marked by the cycle of seasons, gives the year (~365¼ days). These natural cycles are the basis of all calendars.

Q12 A year is about 365¼ days, so a quarter-day is left over each year; adding a leap day every 4 years keeps the calendar aligned with the seasons. Also, 12 lunar months don't exactly equal one solar year, so lunisolar calendars adjust (e.g. with extra months) to keep the Moon's months in step with the solar year.

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